AYUSH SAXENA

Data Scientist | Machine Learning Engineer | AI Engineer

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Summary

MSc Data Science graduate (University of Nottingham) with a Computer Science background specialising in Generative AI and ML Engineering. Experienced in building end-to-end systems, from hybrid RAG pipelines and multi-agent architectures to Deep Learning forecasting systems deployed on AWS. Delivered measurable outcomes with 49% answer relevance improvement in LLM systems, 62.5% latency reduction, 73% recall in risk prediction system and 60% forecasting error reduction in production deployments.

Skills

Programming: Python, C++, R, SQL
GenAI & LLMs: RAG, Multi-Agent Systems, LLM Orchestration, Knowledge Graphs, LangChain, LangGraph, LangSmith, Hybrid Retrieval, Prompt Engineering, Text Embeddings, Semantic Search, Sentence Transformers, Hugging Face Transformers, Ollama (Llama 3)
Machine Learning: Regression, Classification, Clustering, Ensemble Methods, Outlier Handling, Hyperparameter Tuning, Model Evaluation, Bias–Variance Analysis, XGBoost
Deep Learning: PyTorch, Tensorflow, Keras, LSTM, GRU
Databases: Neo4j (AuraDB), PostgreSQL, ChromaDB
Data Processing & Analysis: Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering, Dimensionality Reduction (PCA)
Statistics: Hypothesis testing, statistical analysis, experiment validation, performance metrics
MLOps & Deployment: Data Pipelines, FastAPI, Docker, Git, MLflow, CI/CD (GitHub Actions), AWS (EC2), Streamlit, Render, Pipeline Tracing, Latency Optimization, LLM Evaluation

Experience

Shaivi Systems — **Nov 2023 - Dec 2023**
Machine Learning Intern — *Pune, India*

- Built and evaluated supervised ML models for logistics optimisation using Python, performing preprocessing on noisy operational datasets and reduced prediction error by 5%, reducing operational planning error by 3%.

Formula Student (University of Nottingham Racing) — **2024 – 2025**
Control Team Member (FSAI) — *Nottingham, UK*

- Contributed to the development and validation of throttle control algorithms for an autonomous Formula Student race car, collaborating with perception and control teams to improve system robustness for Formula Silverstone 2025.

Projects

AI Financial Risk Analyst | *Python, RAG, Neo4j, LangGraph, LangSmith*

- Built an AI-powered financial risk analysis system using Hybrid RAG (ChromaDB) and Knowledge Graph (Neo4j AuraDB) to extract and link company risks with mitigation strategies.
- Designed a multi-agent pipeline using LangGraph to dynamically route queries and generate structured financial insights.
- Improved answer relevance by 49% and groundedness by 5.4% over baseline RAG through integration of structured graph reasoning.
- Reduced response latency by 62.5% (80s to 30s) by optimizing model loading, minimizing redundant LLM calls, and improving pipeline efficiency.
- Developed a FastAPI backend and Streamlit UI for real-time query analysis, enabling structured outputs with categorized risks and linked mitigations.
- Integrated LangSmith for observability, enabling end-to-end tracing of agent decisions, debugging, and latency monitoring.

Retail Demand Forecasting Pipeline | *PyTorch, Tensorflow, MLflow, FastAPI, Docker, AWS EC2, PostgreSQL*

- Built an end-to-end deep learning time-series forecasting system using LSTM and GRU architectures in PyTorch and Tensorflow on multivariate retail demand data, where the best model achieved a baseline MSE of 2266 on the test set.
- Engineered temporal features including lag variables and rolling statistics using PostgreSQL, and trained a Random Forest model, reducing error to MSE of 897 resulting in a 60% improvement over the LSTM model.

- Integrated MLflow for experiment tracking and artifact management, ensuring reproducibility and version control of models and preprocessing components.
- Developed a production-ready inference API using FastAPI and containerized the application with Docker, enabling real-time predictions through a REST endpoint.
- Deployed the system on AWS EC2 with a CI/CD pipeline using GitHub Actions, automating Docker build and model deployment to deliver a publicly accessible ML service.

Bankruptcy Risk Prediction System | *Python, Scikit-learn, XGBoost, SMOTE, SHAP*

- Built an end-to-end bankruptcy risk prediction pipeline using XGBoost and Scikit-learn on a financial dataset (6819 samples, 94 features), achieving ROC-AUC = 0.96 and 73% Recall for early identification of distressed firms.
- Designed a robust modeling workflow incorporating feature reduction (94 to 58), class-pos-weight based class balancing, significantly improving minority class detection in an imbalanced financial risk dataset.
- Applied SHAP explainability analysis to uncover key financial distress drivers and optimized classification performance through threshold tuning based on F2 score, improving recall of bankrupt firms while maintaining balanced precision.
- Translated model outputs into risk-management strategies, proposing credit exposure control and financial monitoring frameworks to support data-driven decision making for investors and financial institutions.

Education

University of Nottingham **Sep 2024 - Dec 2025**
MSc Data Science

Cambridge Institute of Technology **May 2020 - May 2024**
Bachelors of Engineering (Computer Science)

Achievements

- **Ranked 1st**, *Train Delay Prediction ML challenge, Achieved the lowest prediction error (lowest MSE) - The University of Nottingham, 2025*
- **Ranked 2nd**, *ECG Signal Processing ML challenge, Achieved the second best accuracy - The University of Nottingham, 2025*
- **Runner-up**, *Programming Tenacity Boot Camp (C) - Cambridge Institute of Technology, 2022*
- **Bronze Medal**, *7th International Karate Championship - 2016*

Certifications

- Microsoft Azure AI Fundamentals (AI-900) - *Microsoft, Jan 2023*
- Statistics and Machine Learning Certificate - *LearnBay, Jul 2024*
- IBM Data Science & AI Certificate- *IBM, Feb 2026*